

SQL Views

COSC 304 – Introduction to Database Systems

Views

A ***view*** is a named query that is defined in the database.

- To the user, a view appears just like any other table and can be present in any SQL query where a table is present.

A view may either be:

- *virtual* – stored query whose results are produced when queried.
- *materialized* – stores the result of a query. These results may need to be refreshed when underlying data changes.

Creating Views

Views are created using the **CREATE VIEW** command:

```
CREATE VIEW viewName [(col1,col2,...,colN)]
AS selectStatement
```

Notes:

- Select statement can be any SQL query and is called the ***defining query*** of the view.
- It is possible to rename the columns when defining the view, otherwise they default to the names produced by the query.

Views Example

Create a view that has only the employees of department `'D2'`:

```
CREATE VIEW empD2
AS SELECT * FROM emp WHERE dno = 'D2';
```

Create a view that only shows the employee number, name, and title:

```
CREATE VIEW staff (Number, Name, Title)
AS SELECT eno, ename, title FROM emp;
```

- The first example is a *horizontal view* because it only contains a subset of the rows.
- The second example is a *vertical view* because it only contains a subset of the columns.

Removing Views

Views are removed using the **DROP VIEW** command:

```
DROP VIEW viewName [RESTRICT|CASCADE]
```

Notes:

- `RESTRICT` will not delete a view if other views are dependent on it.
- `CASCADE` deletes the view and all dependent views.

View Resolution

View resolution translates a query with a view to use base relations.

```
CREATE VIEW staff (Number, Name, Job)
AS SELECT eno, ename, title FROM emp WHERE dno = 'D2';
```

Query on View

```
SELECT Number, Name
FROM staff
WHERE job = 'EE'
ORDER BY Number;
```

Query on Base Relations

```
SELECT eno, ename
FROM emp
WHERE dno = 'D2' AND title = 'EE'
ORDER BY eno;
```

1) **Replace the view:** Substitute its defining query and translate view columns into base-table columns.
2) **Combine the queries:** Incorporate conditions and other SQL clauses to produce an equivalent query on the base tables.

View Updatability

A virtual view automatically reflects changes to its underlying tables. Some views allow updates to the underlying tables through the view.

A view is generally *updatable* if:

- It is defined over a single table (some databases also support certain join views).
- It does not use `DISTINCT`, aggregate functions, `GROUP BY`, or `HAVING`.
- It does not use set operations such as `UNION`.
- Updated columns correspond directly to columns in the underlying table.
- Updates do not violate database integrity constraints.

View WITH CHECK OPTION

WITH CHECK OPTION is used for updatable views to prevent updates through a view that would remove rows from the view.

```
CREATE VIEW staffD2 (Number, Name, Job, DeptNum)
AS SELECT eno, ename, title, dno FROM emp
                              WHERE dno = 'D2';

UPDATE staffD2 SET DeptNum = 'D3' WHERE Number='E3';
```

This update would remove the employee `E3` from the view because changing his department to '`D3`' no longer matches the view which only contains employees in department '`D2`'. To prevent this, can specify the `WITH CHECK OPTION`.

```
CREATE VIEW staffD2 (Number,Name,Job,DeptNum)
AS SELECT eno, ename, title, dno FROM emp WHERE dno = 'D2'
WITH CHECK OPTION;
```

View Updatable Question

Question: Select one view definition that would be updatable.

A)
```
CREATE VIEW v AS
   SELECT DISTINCT salary FROM emp
```

B)
```
CREATE VIEW v AS
   SELECT * FROM emp WHERE eno > 'E3'
```

C)
```
CREATE VIEW v AS
   SELECT dno, AVG(salary) FROM emp GROUP BY dno
```

D)
```
CREATE VIEW v AS
   SELECT eno FROM emp WHERE dno = 'D1' UNION
   SELECT eno FROM emp WHERE dno = 'D2'
```

Advantages and Disadvantages of Views

Advantages:

- Data independence - views can hide some changes in the underlying tables.
- Security - views can be used to limit access to certain data to certain users.
- Easier querying - using views can often reduce the complexity of some queries.
- Convenience/Customization - users only see the parts of the database that they have interest and access to.

Disadvantages:

- Limited updates – complex views may not support INSERT, UPDATE, or DELETE.
- Performance and maintenance – complex virtual views may be expensive to execute; materialized views require storage and refreshing.

Views Practice Questions

Create views that:

- 1) Show only employees that have title '`EE`'.
- 2) List only projects that have someone working on them.
- 3) List only the employees that manage another employee.
- 4) List the department number, name, and average employee salary in the department.
- 5) Show all employees but only lists their number, name, and title.
- 6) Show all employee and `workson` information for employee '`E1`'.

Workson database:

```
emp (eno, ename, bdate, title, salary, supereno, dno)
proj (pno, pname, budget, dno)
dept (dno, dname, mgreno)
workson (eno, pno, resp, hours)
```

Conclusion

Views define virtual relations over base relations. This is useful to:

- reduce query complexity
- improve performance (for materialized views)
- provide different user views of the database
- allow security to be enforced on subsets of the schema

Simple views are often updatable, while more complex views may be read-only.

Objectives

- Define views using `CREATE VIEW` from a high-level description
- Explain the process of view resolution
- Explain when a view is updatable and argue why updating views is not possible in certain cases
- Explain what the `WITH CHECK OPTION` does for views
- List advantages and disadvantages of views


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